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MCSA PROJECT

Active Directory Infrastructure
Implementation

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“1. Introduction

This **report** explains a **Windows Server infrastructure project** that was built to show how a complete Active Directory environment can be set up and managed. The design includes several domain controllers, child domains, and important network services that work together to create a secure and organized system.

The setup includes:

- **Primary** and **Additional Domain** Controllers for redundancy.
- **Child domains** for different branches, such as Ismailia and Alexandria, giving each location its own structure.
- **Read-Only Domain Controller (RODC)** for secure branch authentication.
- Integrated **web hosting (IIS), FTP, DNS services**, and centralized configuration through Group Policy
- **Roaming profiles** and **Remote Desktop** Services to support user mobility and flexible access.
- **Windows deployment service (WDS)** to automate deployment of 50 windows clients.

Overall, this project demonstrates advanced system administration practices, showcasing skills in domain architecture, enterprise service integration, and scalable deployment strategies.

“2. Project Overview

The main goal of this project was to build and configure a Windows Server infrastructure that is secure, scalable, and efficient. The design aims to reflect the structure of a real enterprise environment, complete with multiple branches and centralized administration. By doing so, the project demonstrates how modern organizations can maintain reliability, enforce policies, and expand their systems with minimal effort.

Key Reasons Behind the Design?

This design was chosen to highlight:

- **High Availability:** Multiple domain controllers keep services running.
- **Branch Separation:** Child domains represent different office locations.
- **Security:** RODC ensures safe authentication in less secure sites.
- **Scalability:** Forest structure allows easy expansion.
- **Centralized Control:** Group Policy enforces consistent settings.
- **Automation:** WDS speeds up client deployment.

3. Network setup and infrastructure

DC1 is a primary Domain Controller
DC2 is an additional Domain Controller
DC3 is a RODC, DC4 & DC5 are Child DC

[A@ITI.local](#) can only login to PC1 but can't login to pc1 on Fridays
[help@ITI.local](#) can login to Rodc & his PSWD is replicated to Rodc
[c@ITI.local](#) can't access Flash memory & control Panel & his wallpaper is ITI logo
[A@Ism.ITI.Local](#) can login to PC5-PC1-PC4 (ROMING PROFILE)**

DOMAIN ADMIN need to install **WINRAR** on pc2 using GPO (how)**

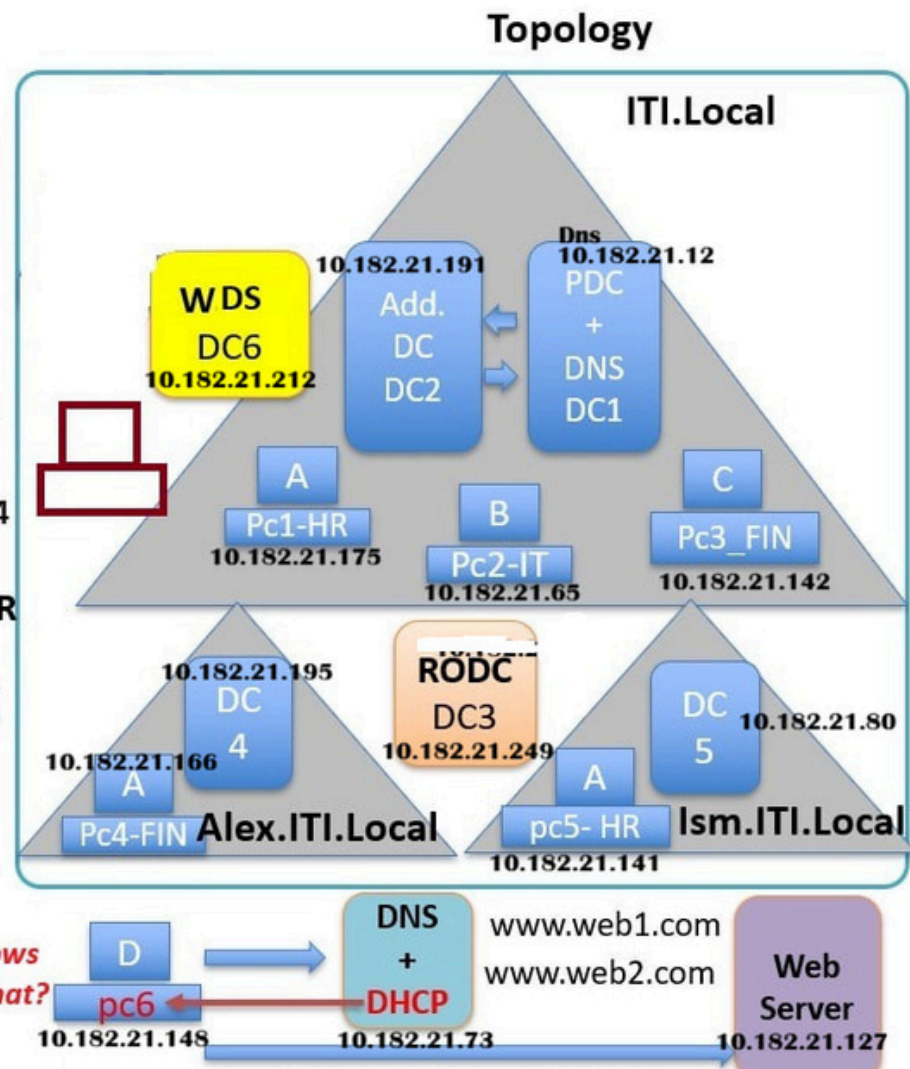
DOMAIN ADMIN delegate to [B@iti.local](#) to login remotely to DC1 (not member of administrators)**

[A@ITI.local](#) check the website

<https://www.web2.com> from pc1

you need to add 50 new computers with windows and 50 user to the domain (how to automate that?)
WDS

D is a local user on **pc6** but he can manage remotely (**RDP**) the webserver with administrative privileges, his responsibilities is to check <http://www.web1.com> and get a **copy** of it using **FTP**



- IP addresses are demonstrated in the figure.
- Domain structure: ITI.Local.
- Domain controllers:
 - DC1 → primary
 - DC2 → additional
 - DC3 → RODC
 - DC4 & DC5 → Child
 - DC6 → WDS server

1. Unified Network Creation:

All devices connected via mobile hotspot to create isolated lab environment.

2. IP Address Planning:

Designed IP addressing scheme for servers and clients

3. Static IP Assignment: Configured static IPs on all physical PCs and virtual machines

4. DNS Configuration: Set primary and secondary DNS to point to domain controllers

5. Network Validation: Tested connectivity between all devices

Reason for using Static IP Addresses?

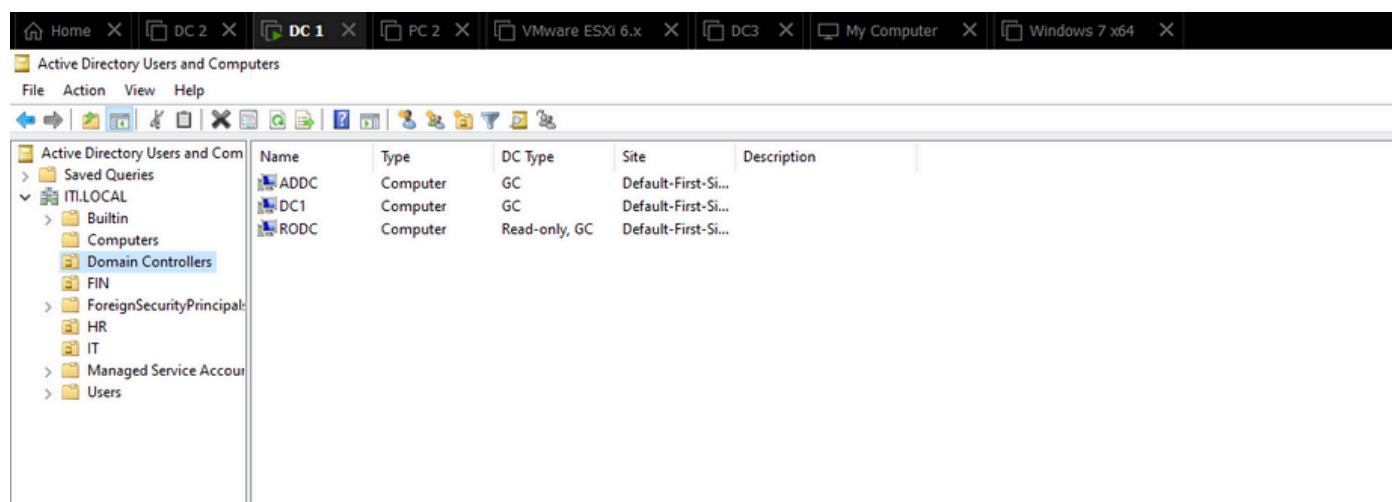
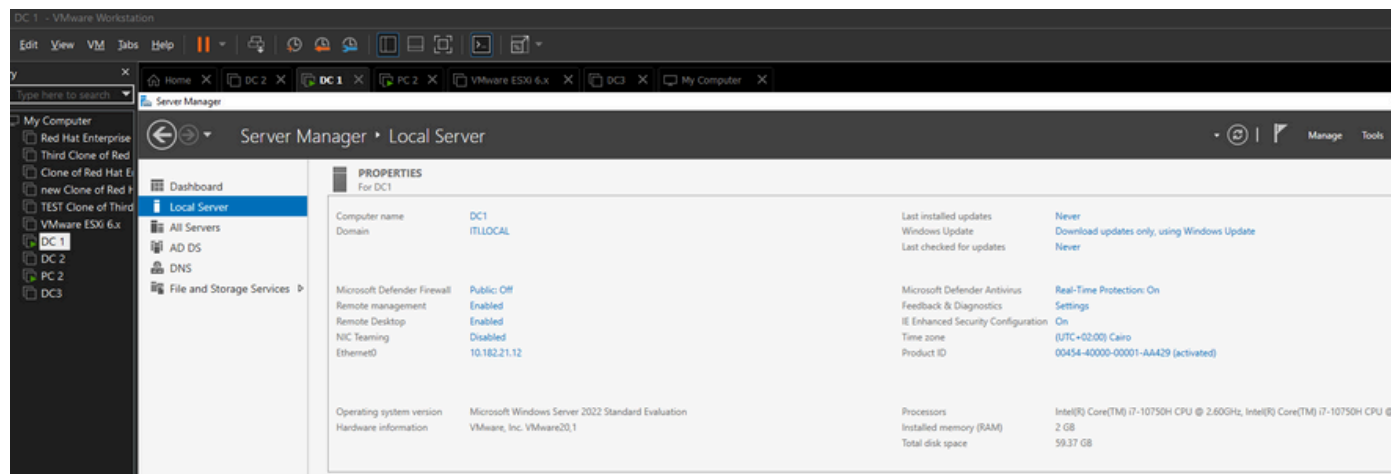
Static IP addresses were assigned to all servers and critical infrastructure components for several important reasons:

- **DNS Reliability:** Domain Controllers act as DNS servers and require consistent IP addresses
- **Service Stability:** Prevents service disruption due to IP changes
- **Active Directory Requirements:** AD DS depends on stable network addressing
- **Simplified Troubleshooting:** Consistent addressing makes network issues easier to diagnose
- **Client Configuration:** Clients point to specific DC IP addresses for authentication

4. Core Domain controllers Deployment

4.1. Primary Domain Controller “DC1”

The Primary Domain Controller serves as the foundation of the Active Directory forest. It hosts the initial domain (ITI.local) and provides core services such as authentication, authorization, and DNS integration. By holding the first copy of the Active Directory database, DC1 ensures stability and consistency across the environment. It is also responsible for replicating directory data to additional domain controllers, making it the central point of trust and management in the infrastructure.



4.2. Additional Domain Controller (ADC) “DC2”

The **ADC** is a critical component in the Active Directory infrastructure, designed to ensure **high availability** and load balancing.

In this project, an additional server named **ADDC** was configured to support the Primary Domain Controller (DC1).

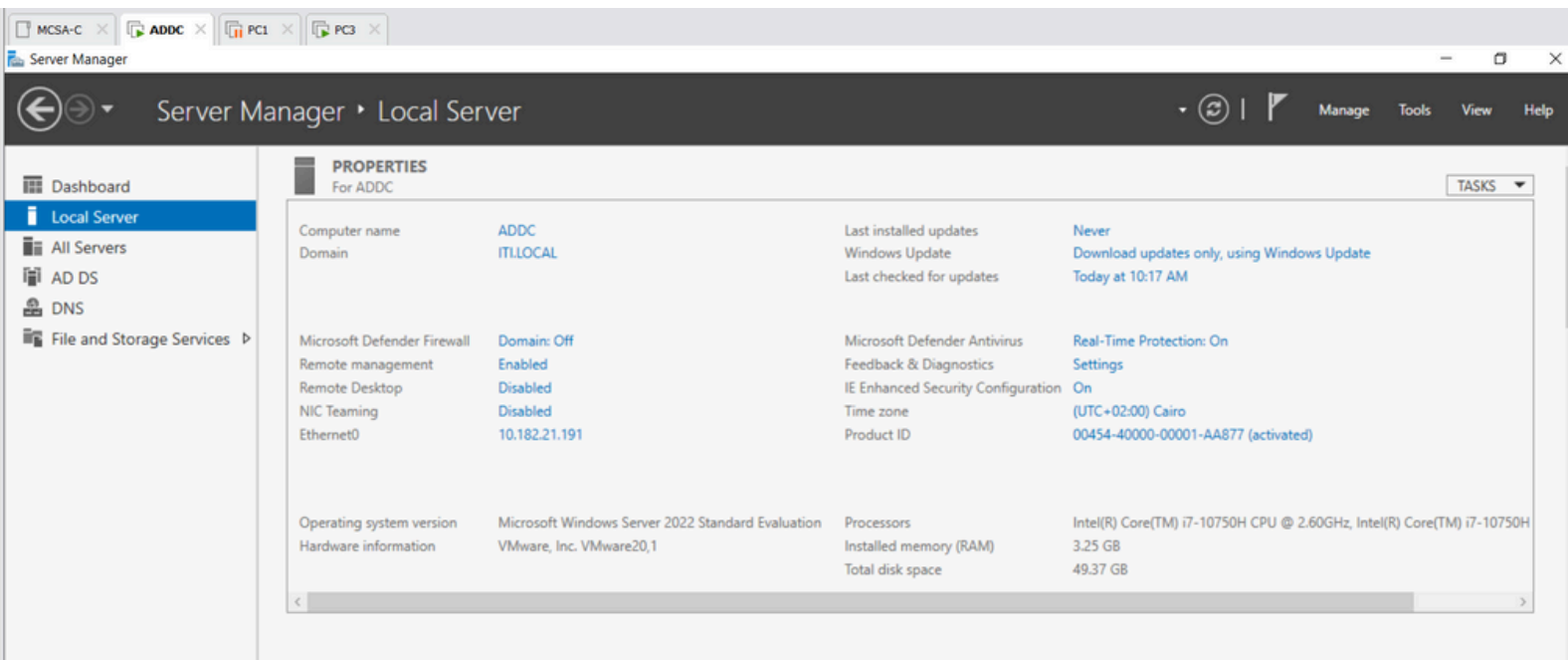
Technical Importance of ADC in the Network:

High Availability: It ensures domain services remain available if the primary controller (DC1) fails, preventing authentication downtime for users and services.

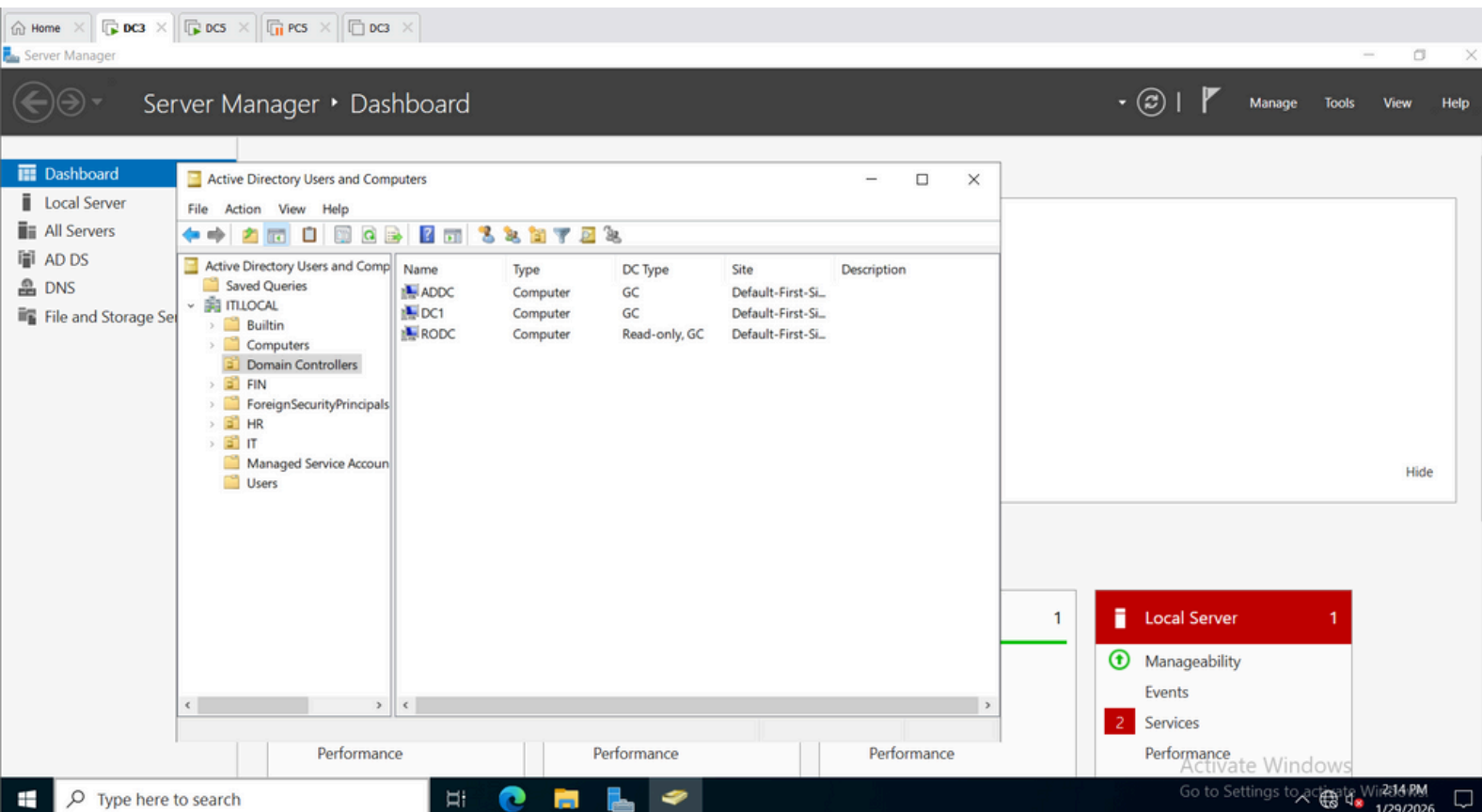
Load Balancing: It helps distribute login requests and directory queries from client machines, reducing the load on the primary server and improving overall network performance.

Disaster Recovery: It maintains a live, replicated copy of the Active Directory database, making it easier to recover data in the event of a total primary server failure.

Feature	Configuration Detail
Server Name	ADDC
Domain	ITI.LOCAL
IPv4 Address	10.182.21.191
Operating System	Windows Server 2022 Standard Evaluation



It maintains a replicated copy of the Active Directory database from the primary one.



“ 5. Child domain controller

Child domain controllers are deployed to extend the Active Directory forest into separate organizational units while maintaining trust relationships with the parent domain.

They provide several advantages:

- **Administrative Delegation:** Local IT teams can manage users and resources independently without affecting the parent domain.
- **Policy Isolation:** Different Group Policies can be applied to each branch, ensuring tailored configurations.
- **Namespace Organization:** Each child domain has its own DNS namespace, making the structure logical and scalable.
- **Performance Optimization:** Authentication requests are handled locally, reducing latency for branch users.

By introducing child domains, the infrastructure gains flexibility and scalability while preserving centralized oversight from the parent domain.

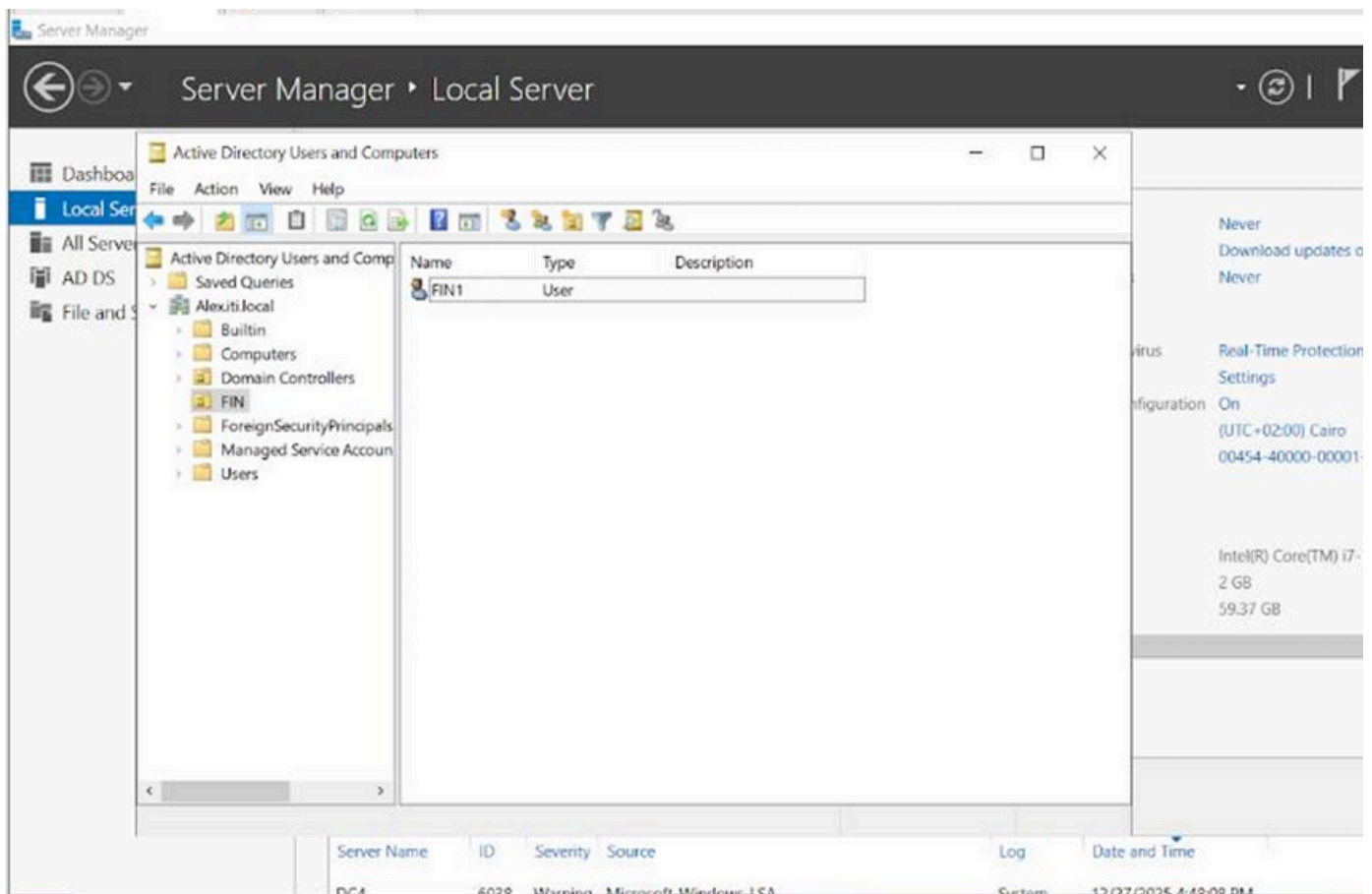
5.1 Alex branch “DC4”

The Alex.ITI.Local child domain was established to represent the Alexandria office within the ITI.Local forest.

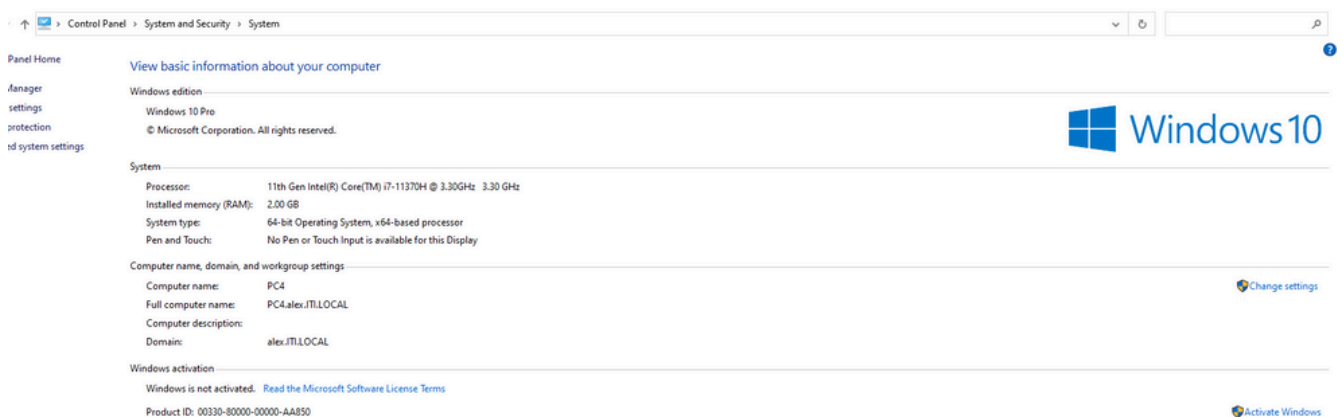
Its configuration included:

- **Domain Setup:** A dedicated server was prepared with static IP addressing and DNS integration pointing back to the parent domain.
- **Organizational Units:** Separate OUs were designed for Alexandria’s departments, enabling branch-specific policies.
- **User and Computer Accounts:** Users for the Alexandria office were added, and PC4 was successfully joined to the Alex.ITI.Local domain.
- **Administrative Control:** Local administrators were delegated limited rights to manage branch resources without impacting the parent domain.

Alex Domain with Fin departement and users:



PC4 joined the domain:

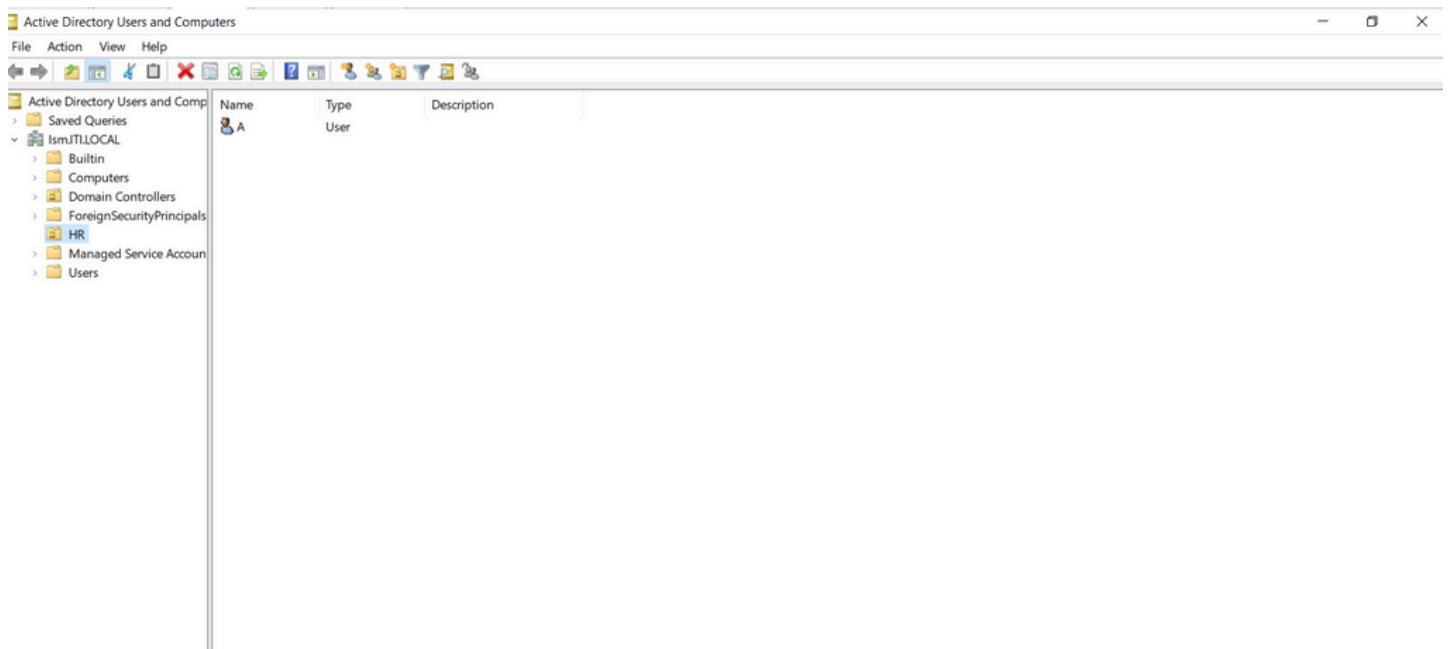


5.2. Ismailia Branch “DC5”

The Ismailia branch requires local administrative control and optimized authentication performance.

Ismailia Domain Configuration:

1. **Server Setup:** Configured server with static IP
2. **AD DS Installation:** Installed Active Directory Domain Services
3. **Child Domain Creation:** Created ism.ITI.local as child of iti.local
4. **Trust Relationships:** Automatic two-way transitive trust with parent domain
5. **DNS Integration:** Configured DNS delegation for child domain
6. **Organizational Units:** Created OU structure for Ismailia users and computers.



6. Read-Only Domain Controller (RODC - DC3)

The **RODC** was implemented to provide domain services to the branch office while maintaining a high security posture.

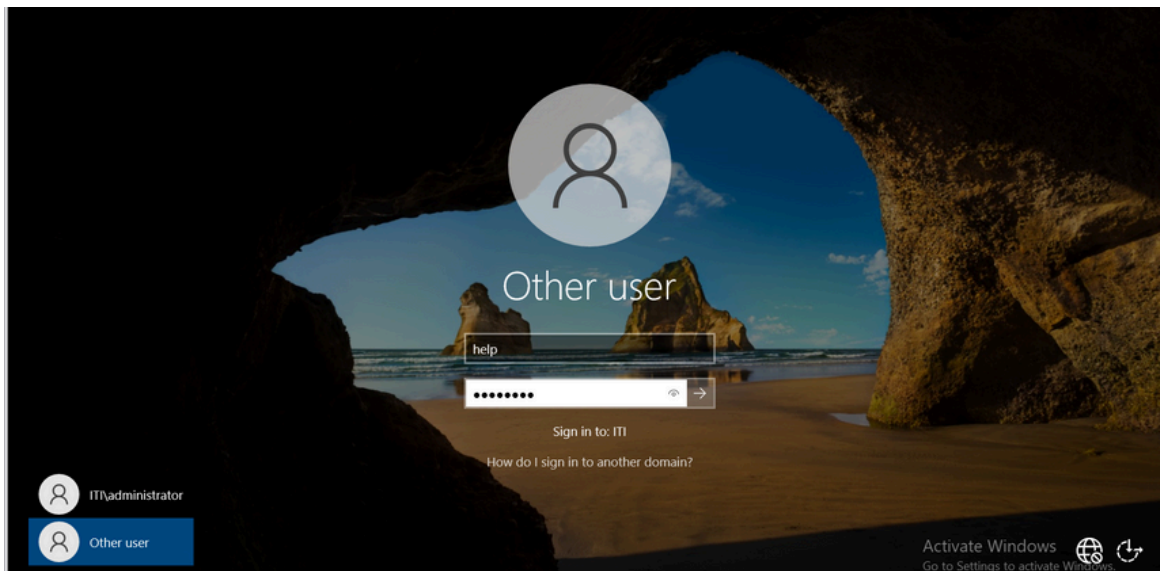
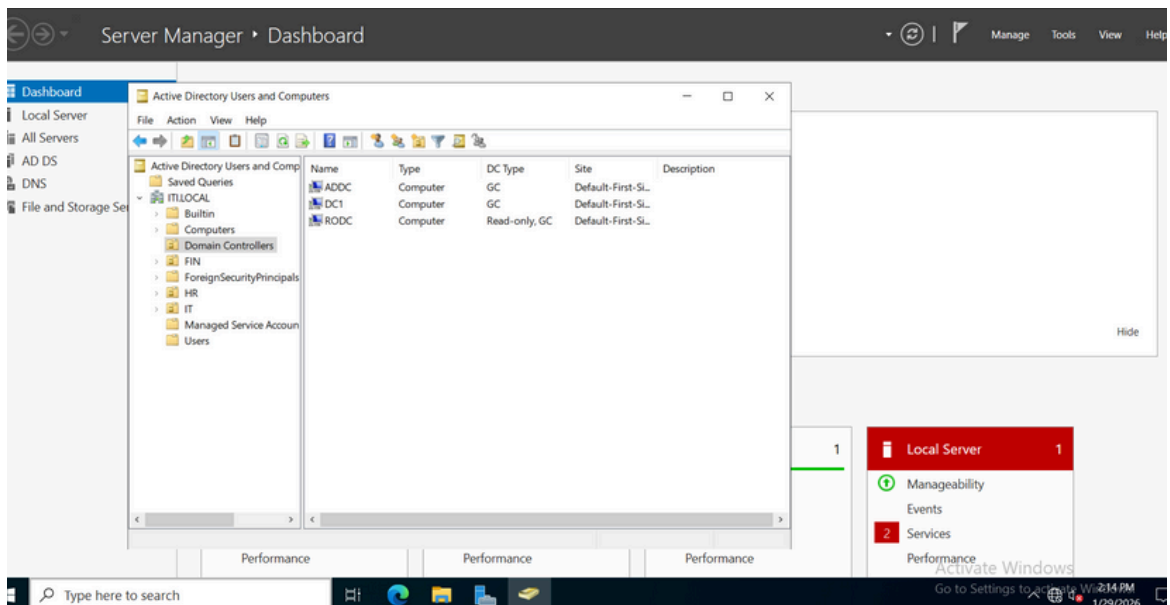
Technical Importance:

Enhanced Security: Since the RODC does not store all user credentials by default, a physical breach of this server does not compromise the entire domain's security.

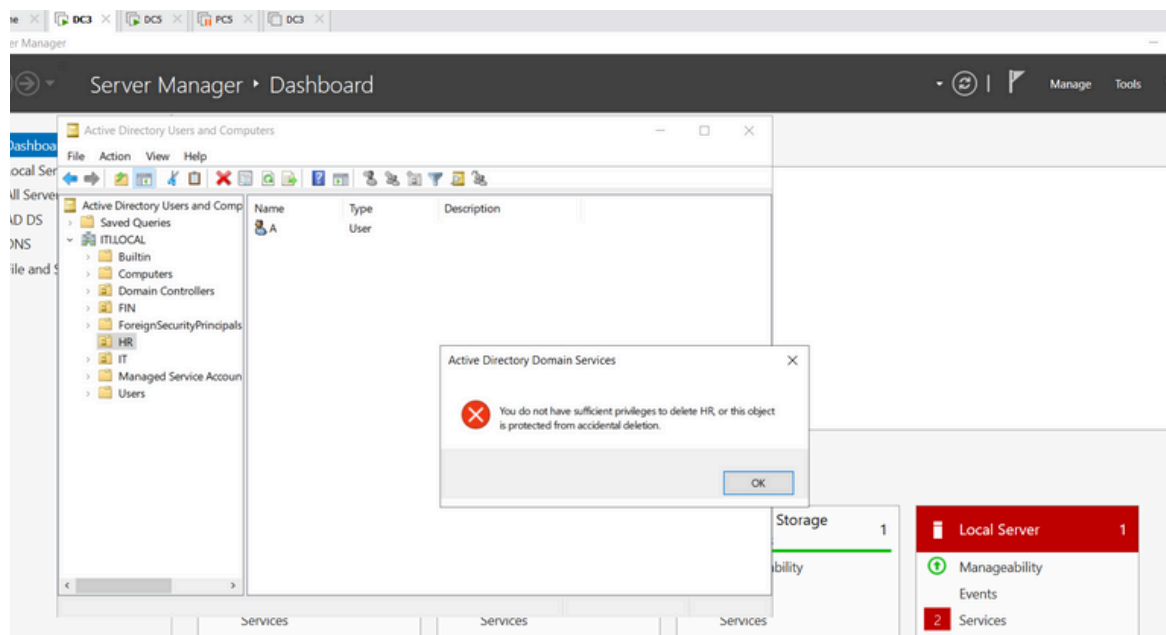
Read-Only Database: The Active Directory database is read-only, preventing unauthorized local changes from replicating back to the primary controllers.

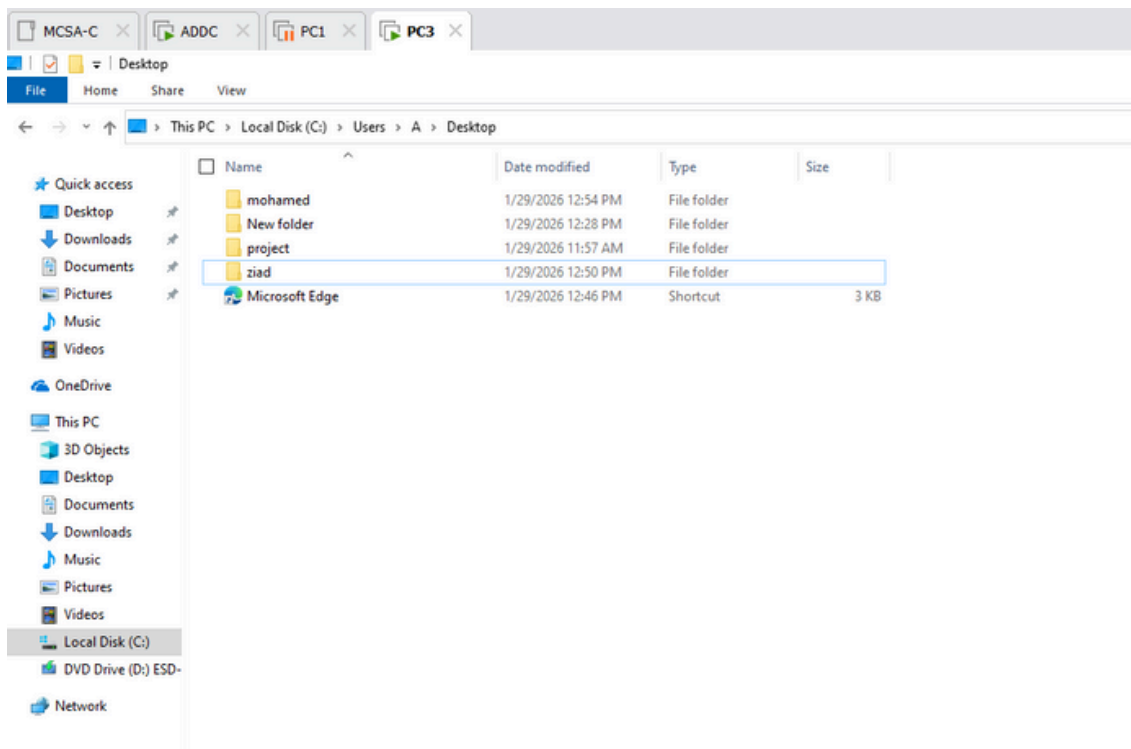
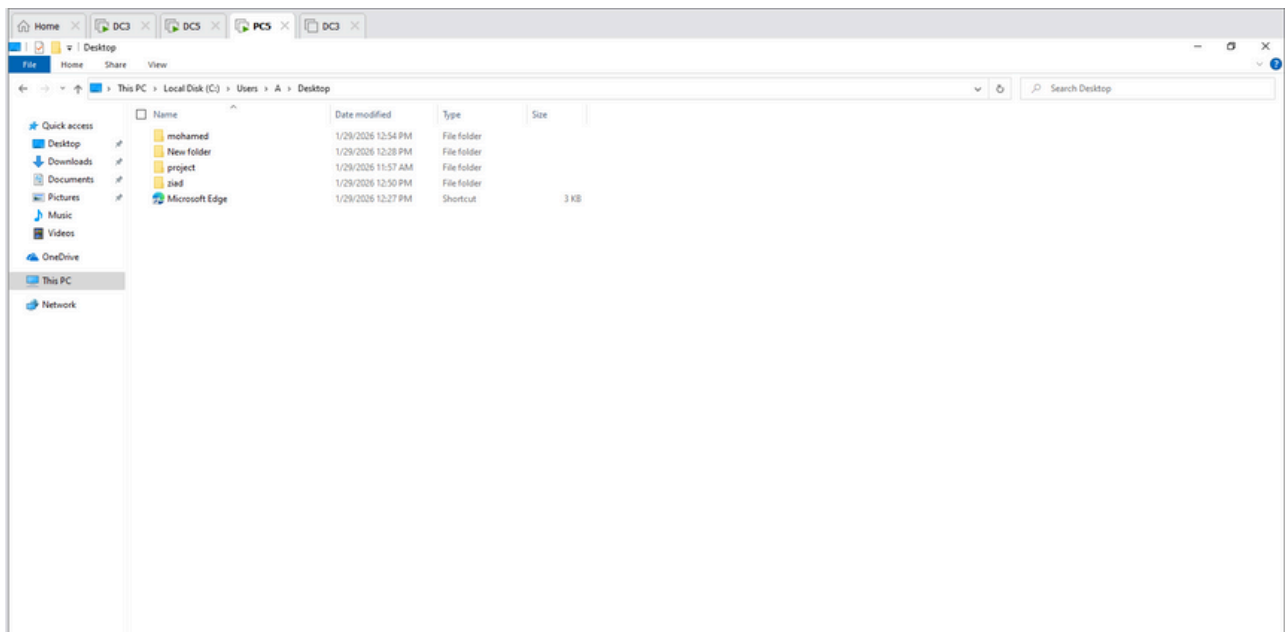
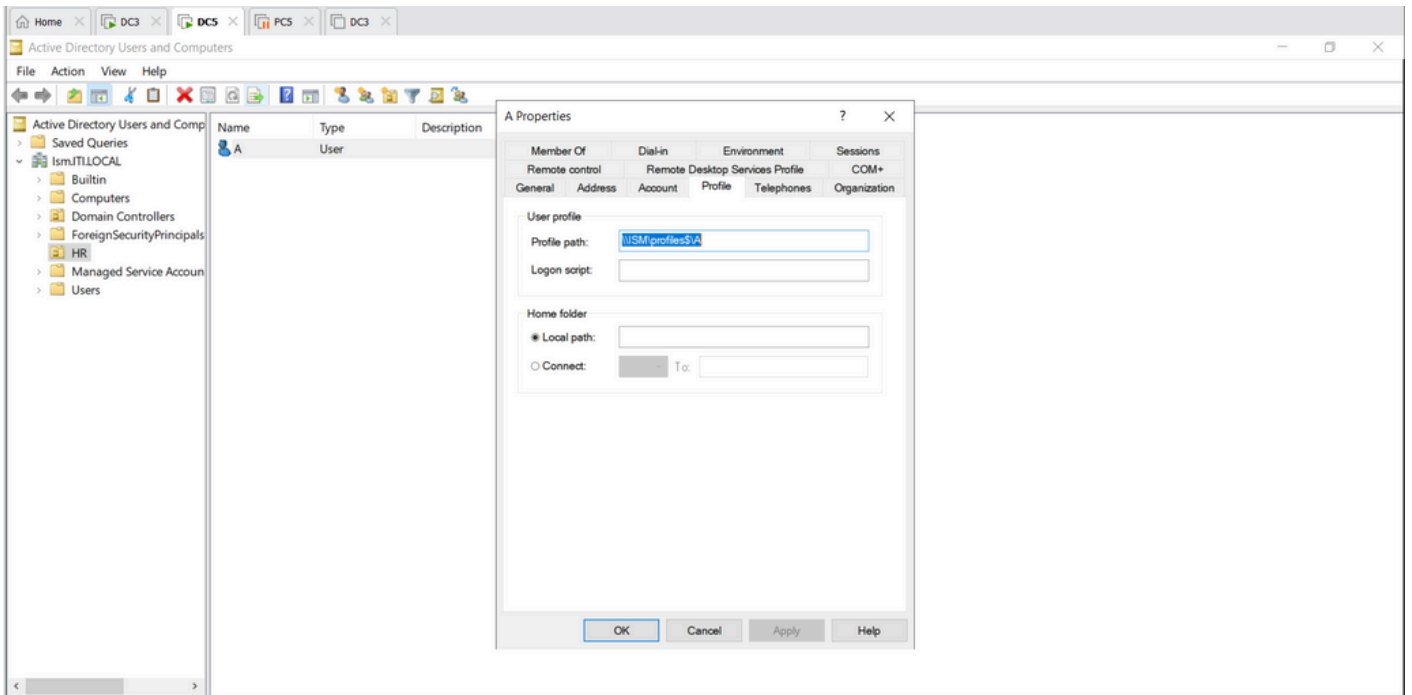
Password Replication Policy (PRP): Allows specific users, such as help@ITI.local, to authenticate locally even if the link to the main site is down.

Feature	Configuration Detail
Server Name	DC3
IPv4 Address	10.182.21.249
Specific Policy	Password Replication allowed for help@ITI.local



verifies it's a read only privilege





“ 7. IIS

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IIS was deployed to provide web hosting capabilities within the organization.

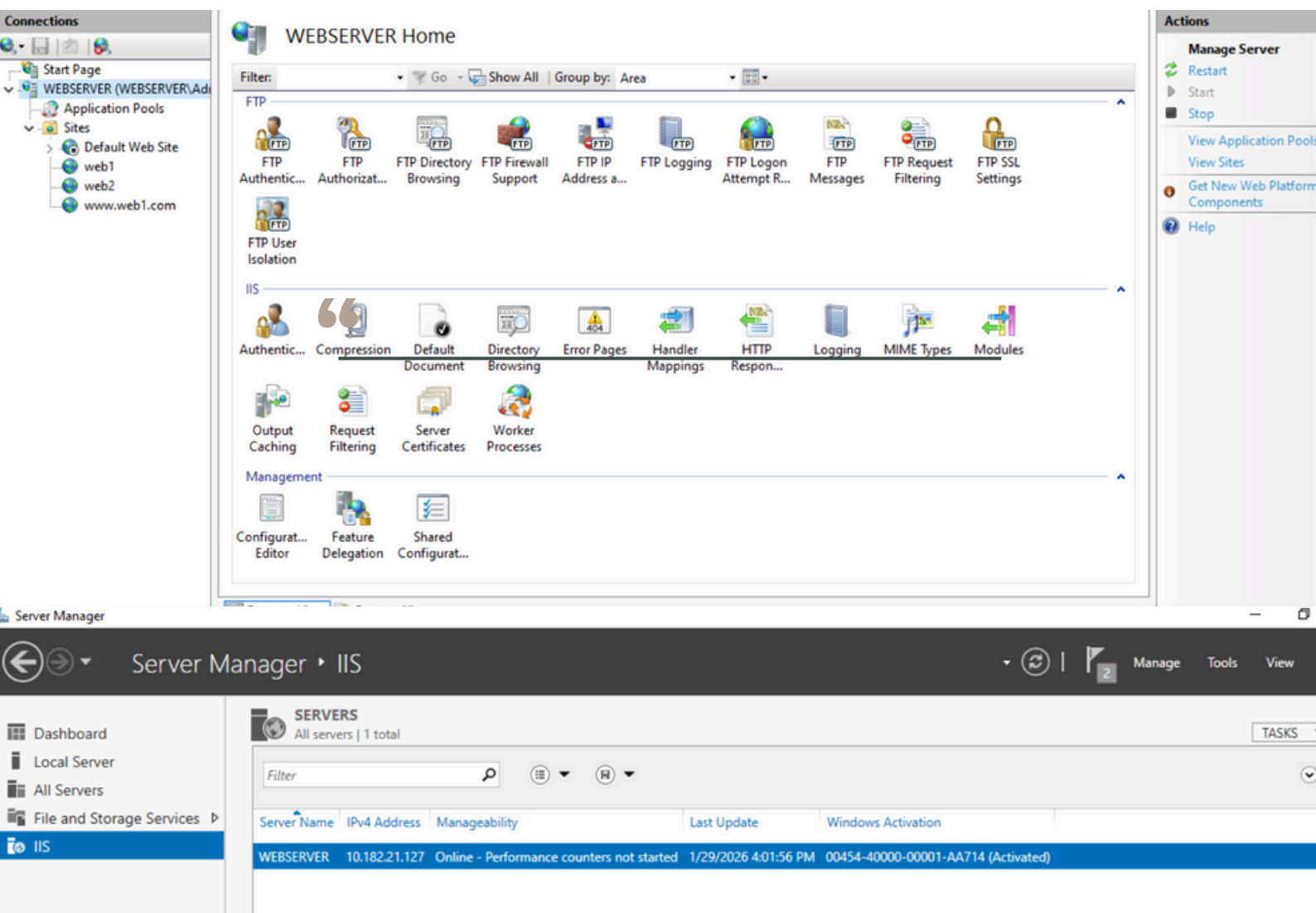
IIS Configuration

1. Role Installation: Added IIS role through Server Manager

2. Feature Selection: Installed required IIS features.

3. Default Website: Configured and tested default website

4. Testing: Verified web service accessibility and functionality



DNS Service Benefits

- **Easy access using domain names:**

DNS allows users to access the web server using a human-readable domain name instead of an IP address, improving usability and accessibility.

- **Centralized name management:**

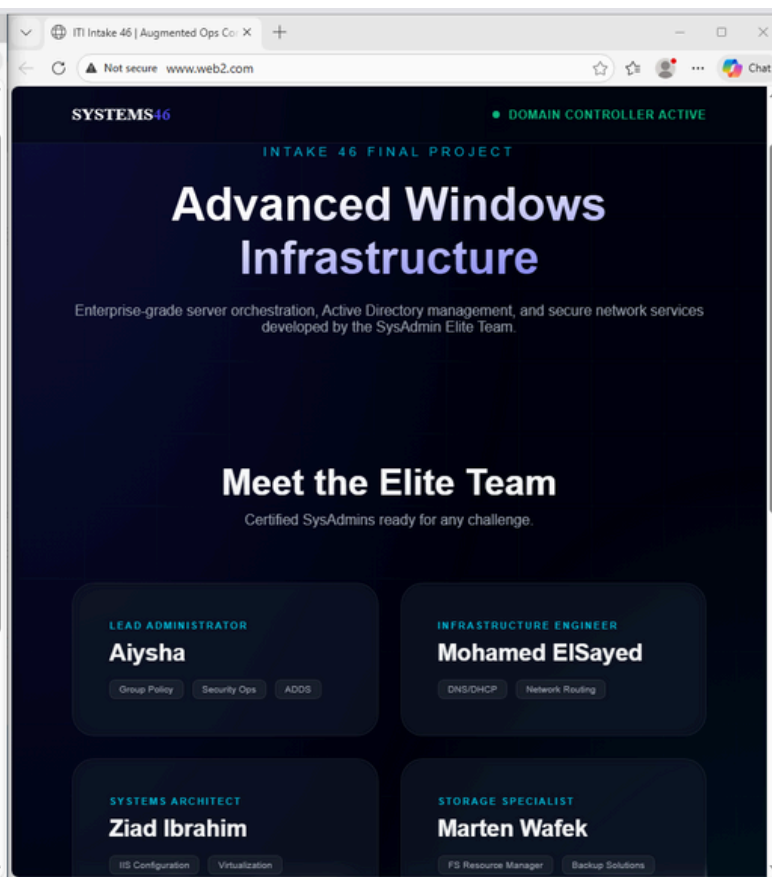
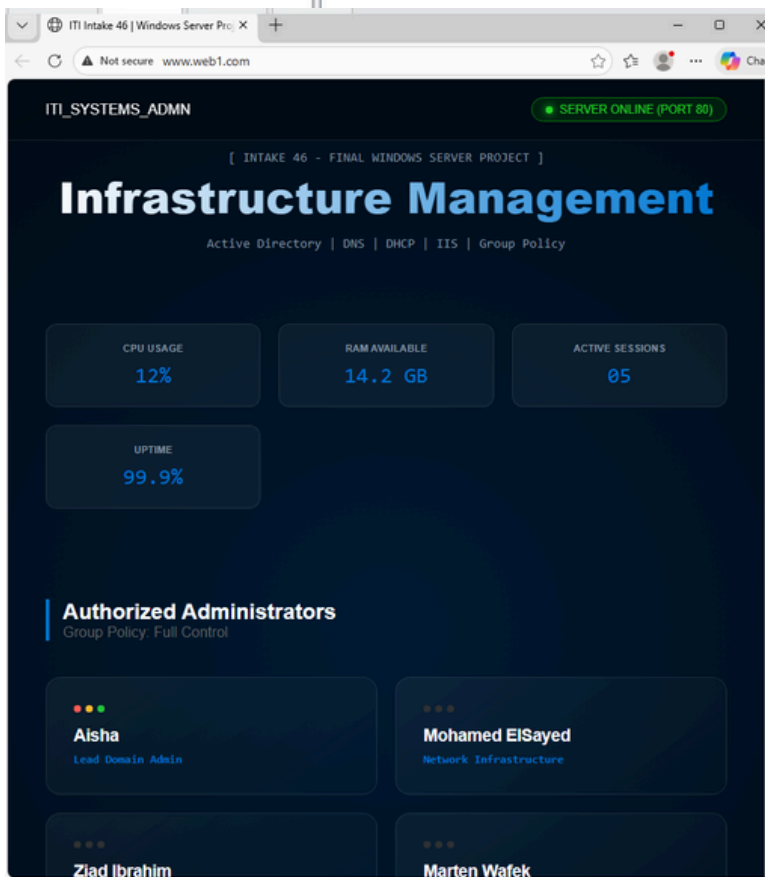
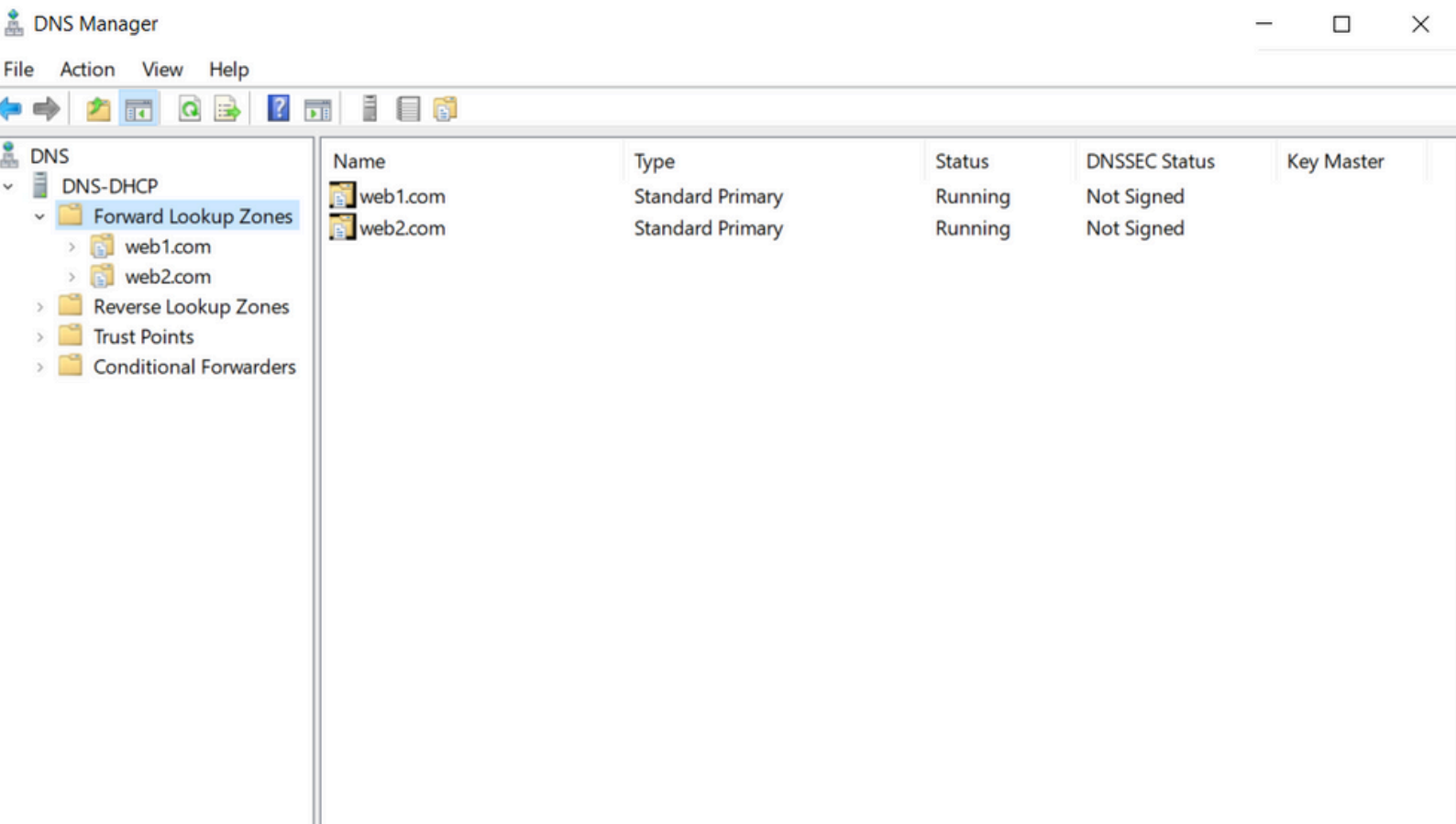
DNS provides a single point of control for managing domain names and IP address mappings, making updates and changes easier to manage.

- **Supports scalability and expansion:**

DNS enables the addition of new servers or services without impacting end users, supporting future growth and load distribution.

- **Active Directory integration:**

DNS integrates seamlessly with Active Directory in Windows Server environments, ensuring reliable service discovery and domain functionality.

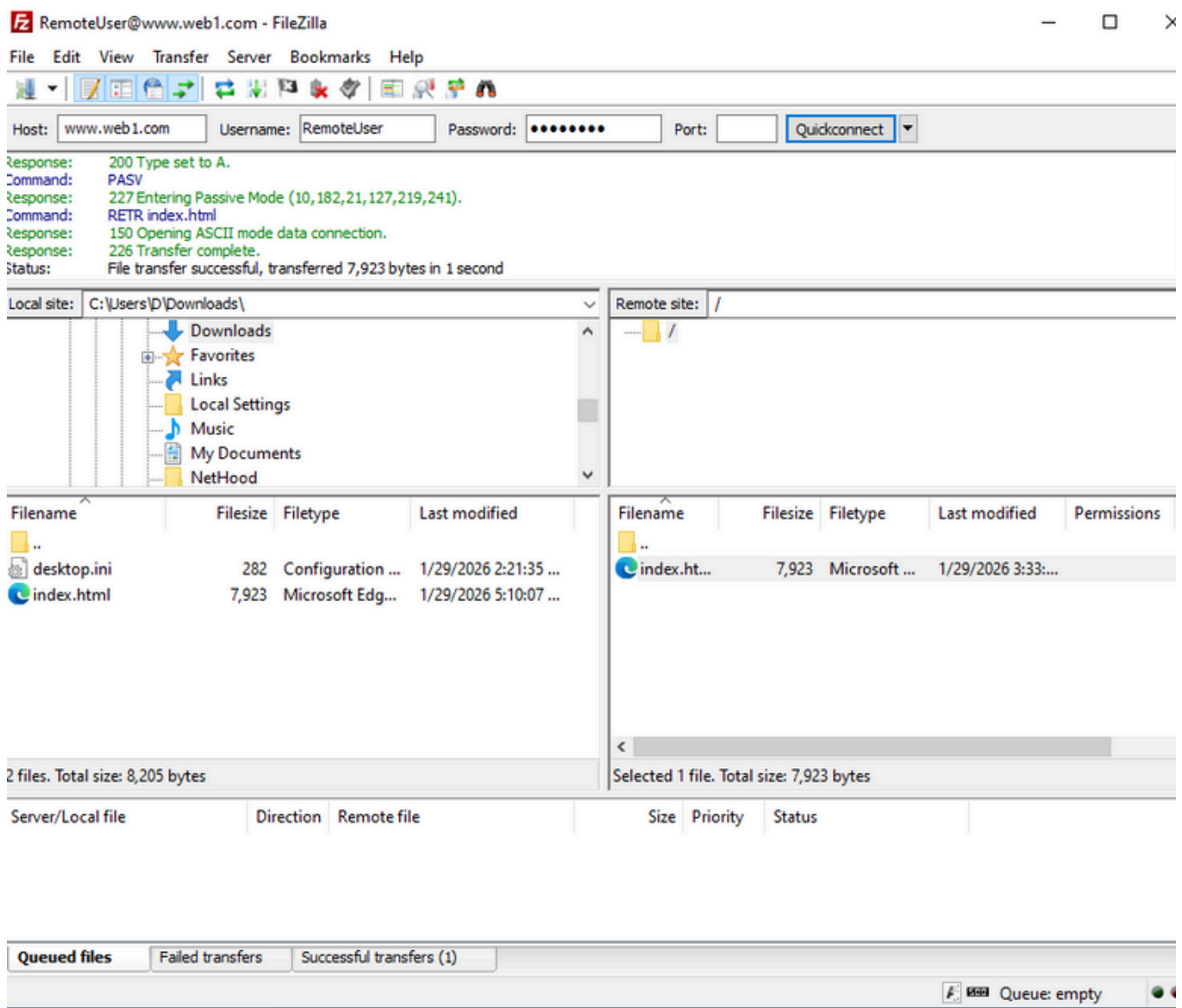


Purpose of FTP Services

- **File Distribution:** Centralized file sharing for large files
- **Remote Access:** Employees can upload/download files remotely

FTP Configuration

1. **FTP Role Installation:** Added FTP Server role to IIS
2. **FTP Site Creation:** Configured FTP site with appropriate directories
3. **Authentication:** Enabled Basic Authentication with AD credentials

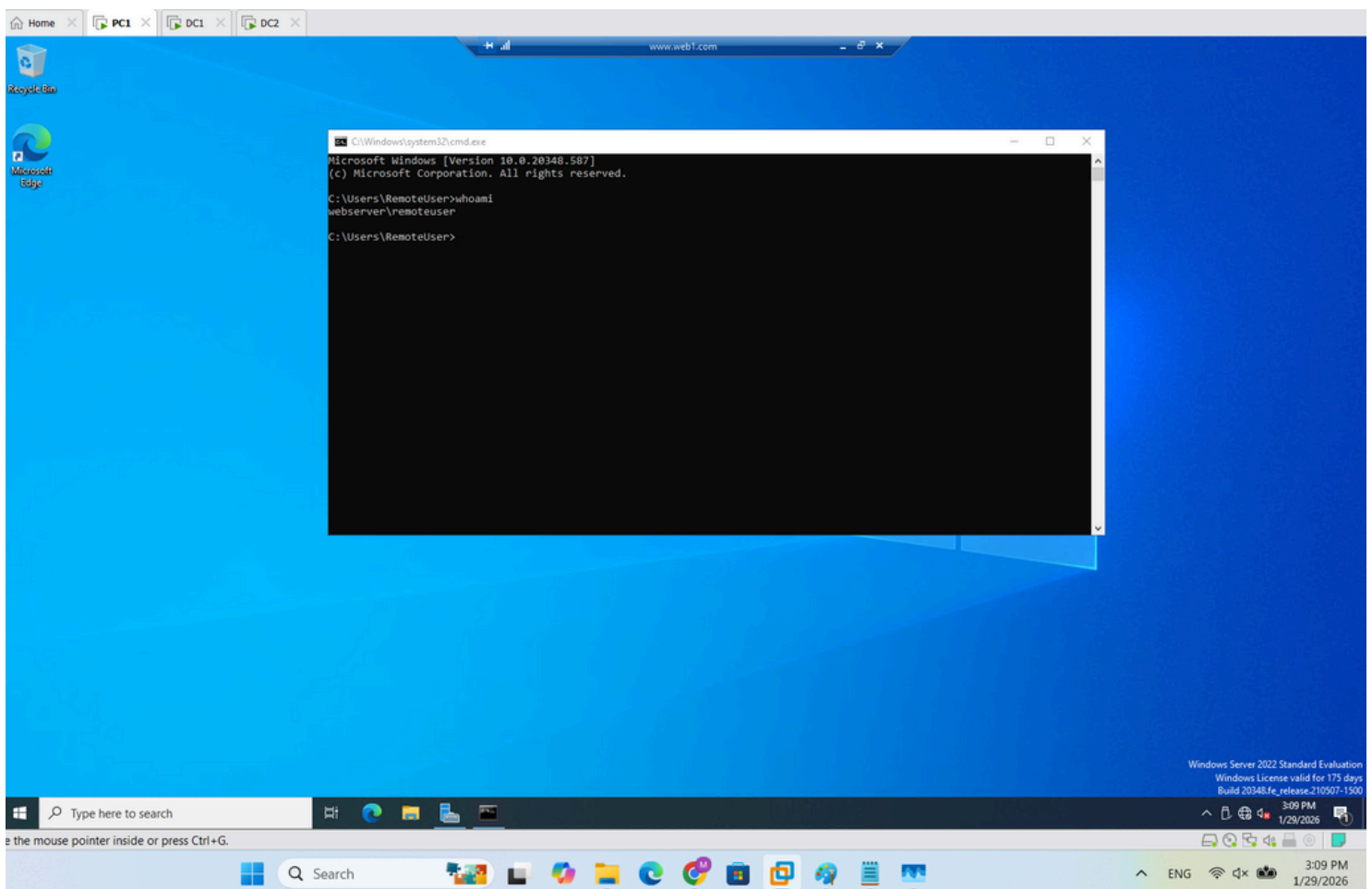


Remote Desktop Services (RDS)

- was configured to provide secure remote access to workstations and enable IT support capabilities.

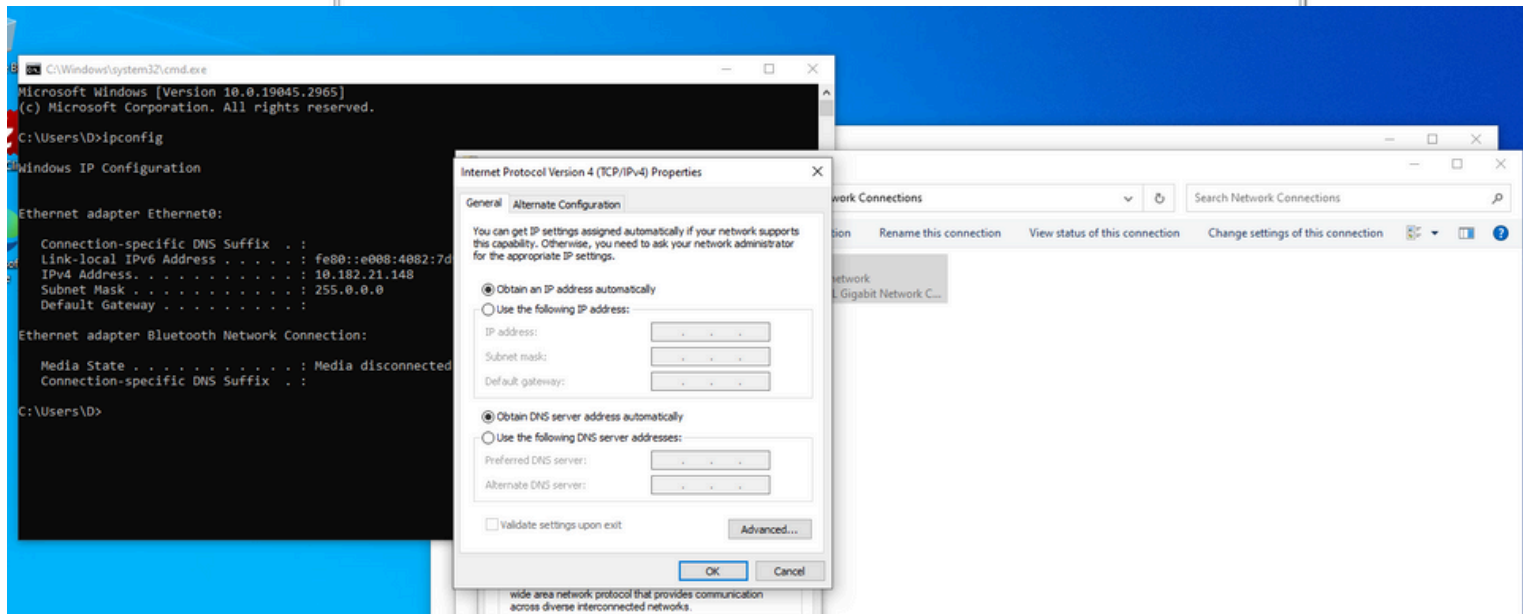
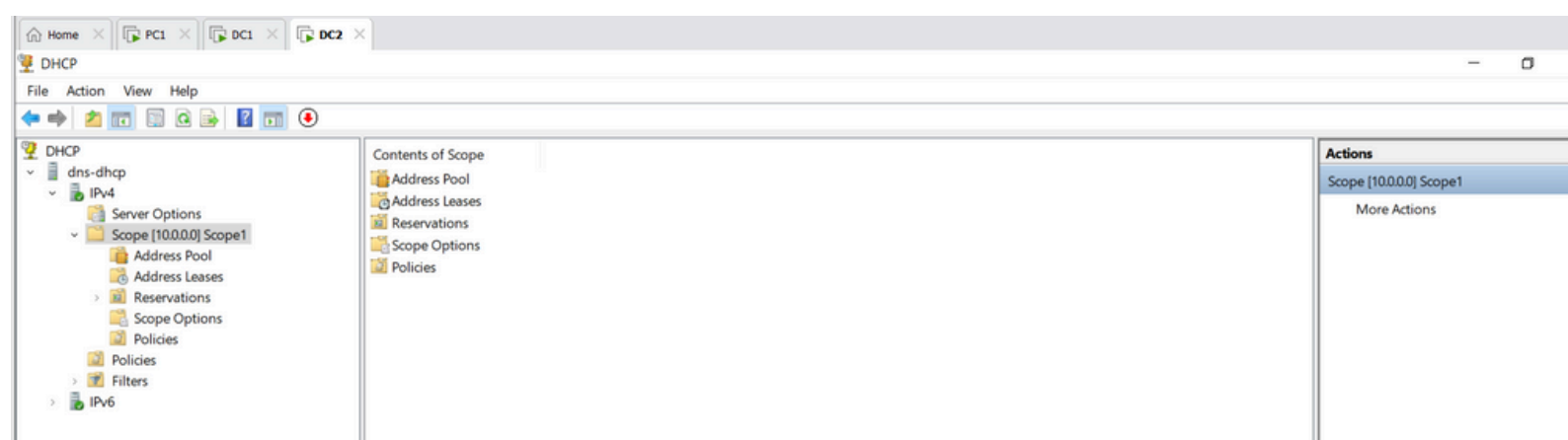
Benefits of Remote Desktop Services

- Remote Work: Users can access their workstations from home or remote locations
- IT Support: Help desk can remotely troubleshoot user issues without physical access
- Centralized Applications: Run applications on servers rather than client machines



Benefits of DHCP Services

- Automatically assigns IP addresses to devices
- Saves time and manual configuration
- Reduces IP address conflicts
- Centralized IP address management
- Easy connection for new devices
- Provides DNS and gateway automatically
- Suitable for large networks

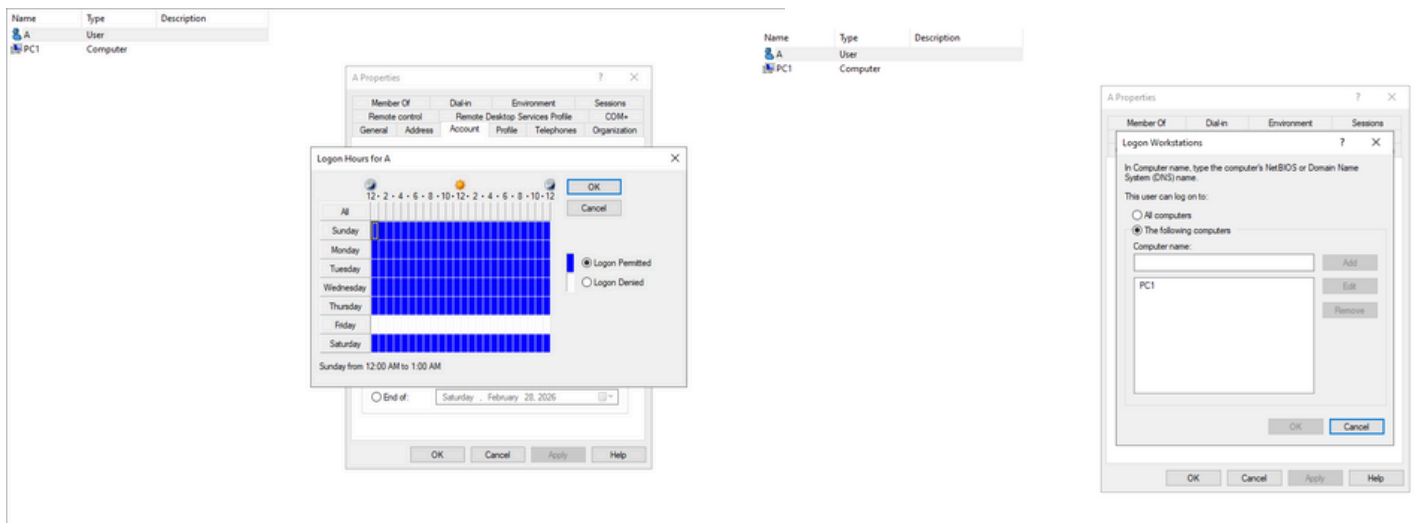


“12. Group policy management

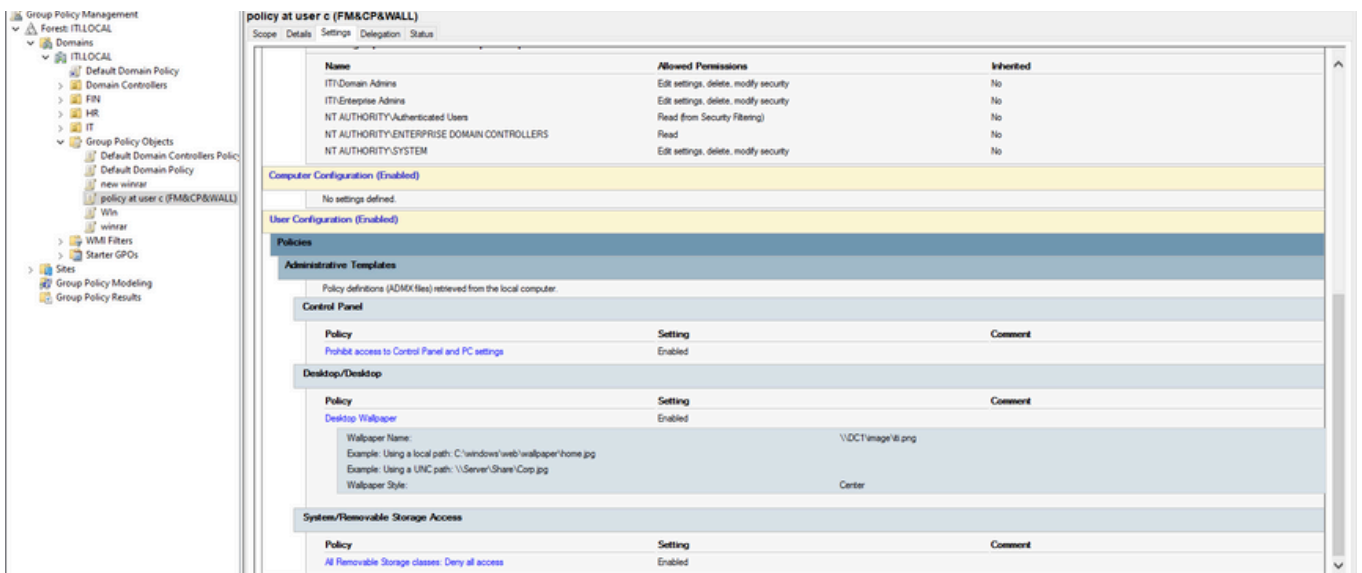
Group Policy provides centralized control over user and computer settings within the domain. Through Group Policy Objects (GPOs), administrators can enforce security rules, configure software, and standardize system behavior across all branches. This ensures consistency, reduces manual configuration, and strengthens overall security in the enterprise environment.

GPOs are utilized for the following purposes:

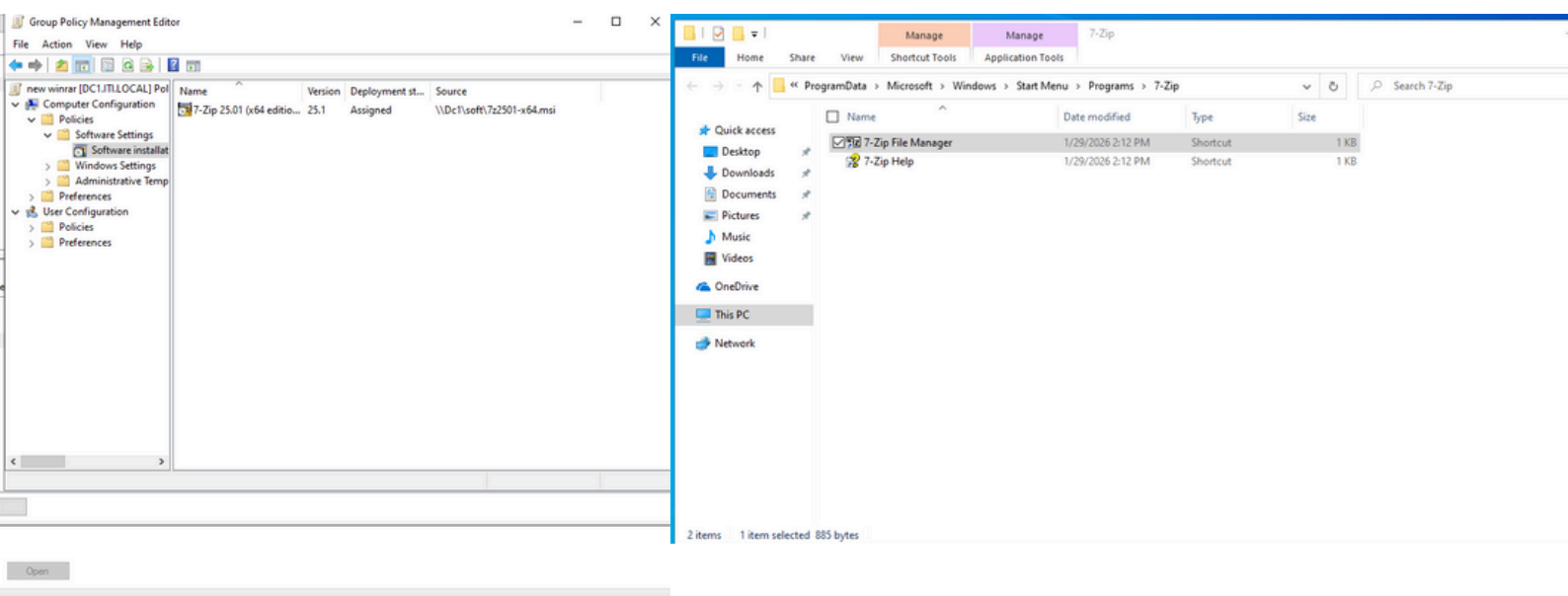
- user A can't logon friday and login in PC1 only:



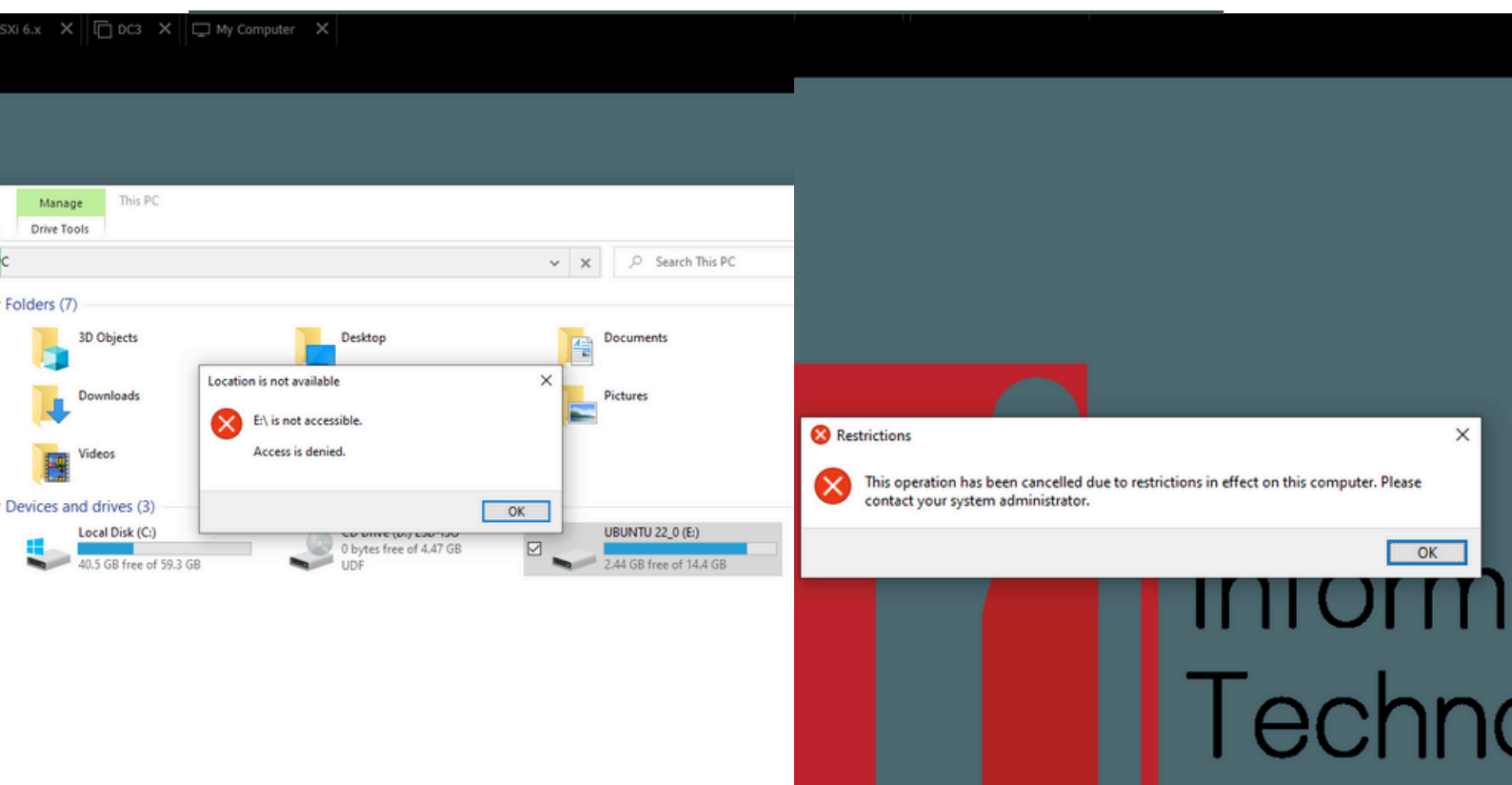
- policies in user C:



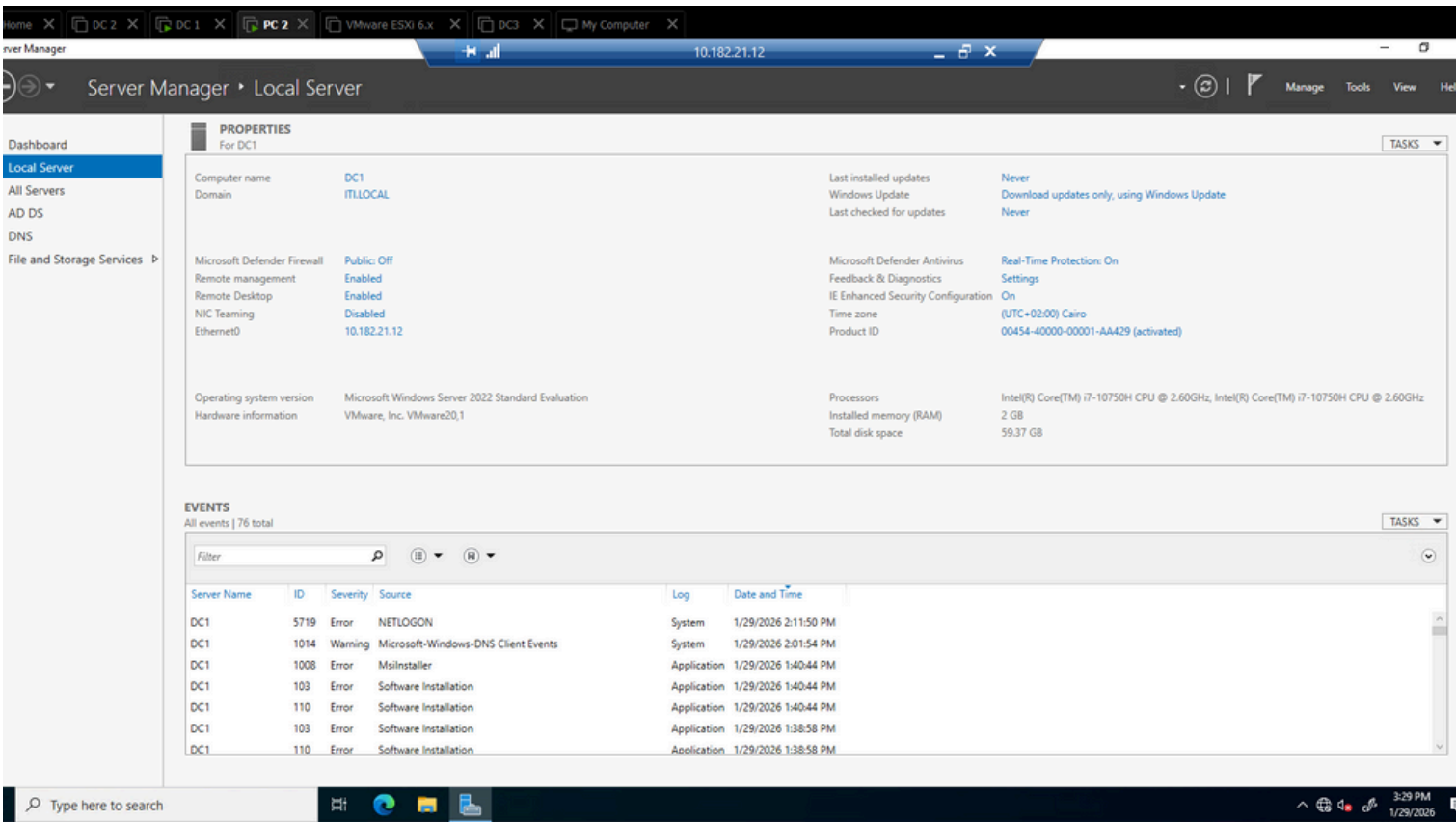
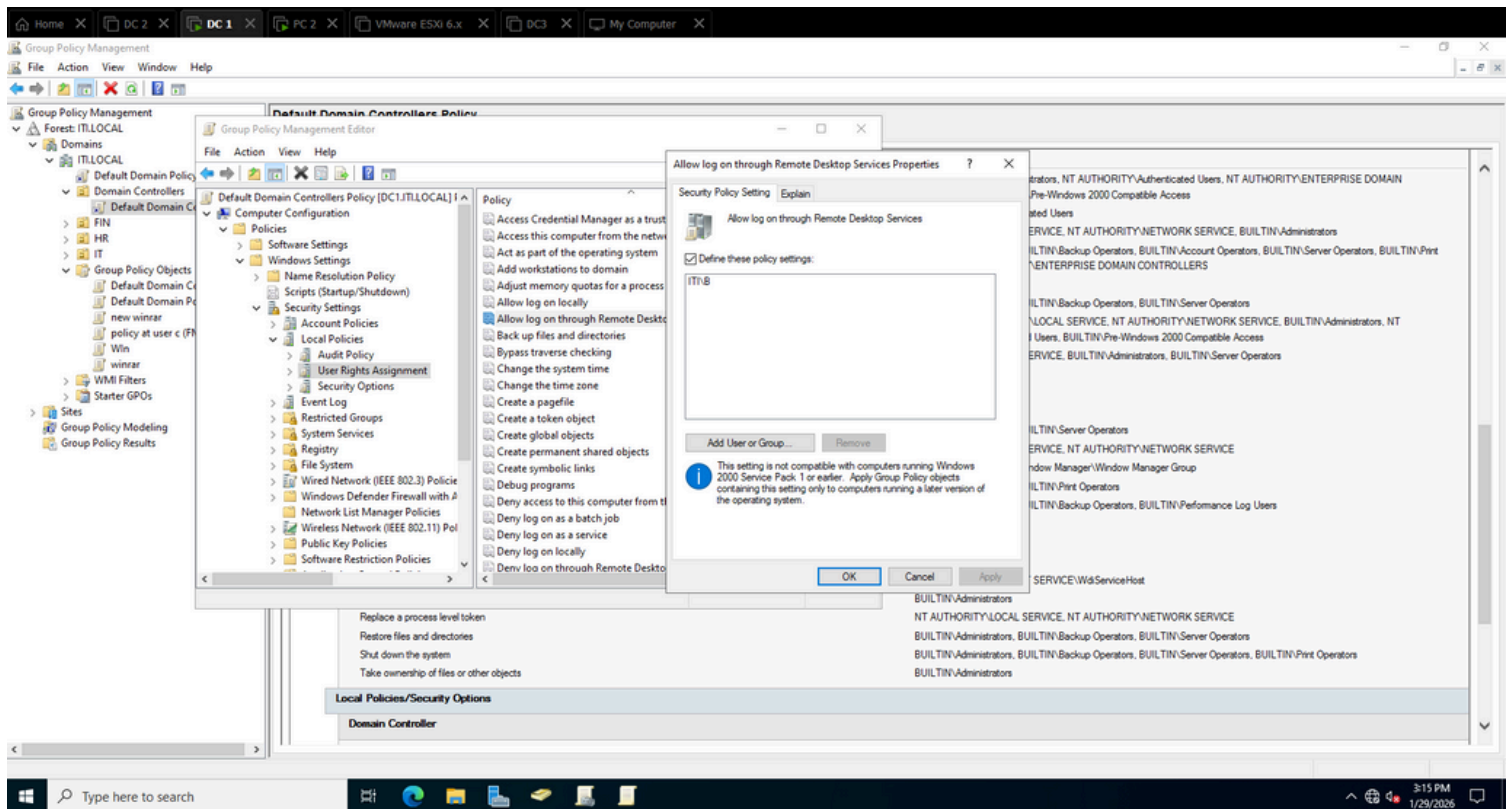
- WinRAR was deployed via the .msi package using Group Policy, allowing silent installation across domain clients for consistency and reduced manual effort:



- Restricted access to USB ports, Control Panel, and desktop wallpaper changes:



- Policy for delegation user B remotely to DC1



13. Roaming User Profiles

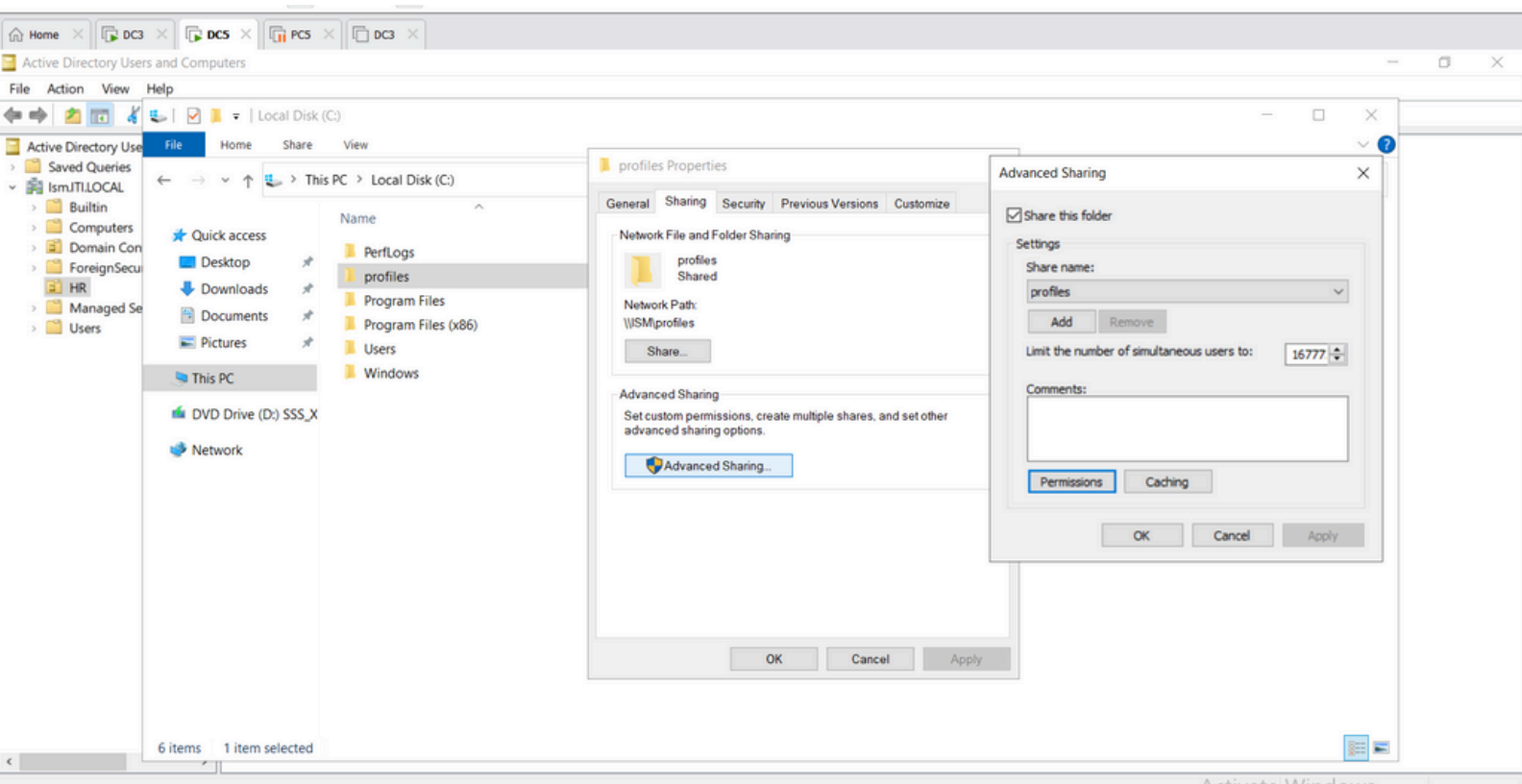
This configuration ensures that user **A@ism.ITI.Local** has a consistent work environment regardless of which physical machine they use (PC5, PC1, or PC4).

Implementation Steps:

- **Centralized Storage:** A shared folder was created on the server with appropriate NTFS and Share permissions.
- **Profile Path Configuration:** The user's profile path in Active Directory was set to the network share (e.g., \\Server\Profiles\%username%).
- **Mobility Testing:** Verified that the user's desktop, documents, and settings persist when moving between PC5-HR, PC1-HR, and PC4-FIN.

Verification:

- Successful login on PC1 (Primary Domain) and PC5 (Child Domain) using the same profile.



“ 14. Windows deployment service (WDS) “DC6”

Windows Deployment Services

was implemented as a **stand-alone server** to streamline the process of deploying operating systems across multiple client machines. To ensure full functionality, the WDS server was configured with the DHCP role, enabling it to assign IP addresses to devices during the boot process. This setup allows computers to boot directly from the network using PXE (Preboot Execution Environment), eliminating the need for manual installation via physical media like USBs or DVDs.

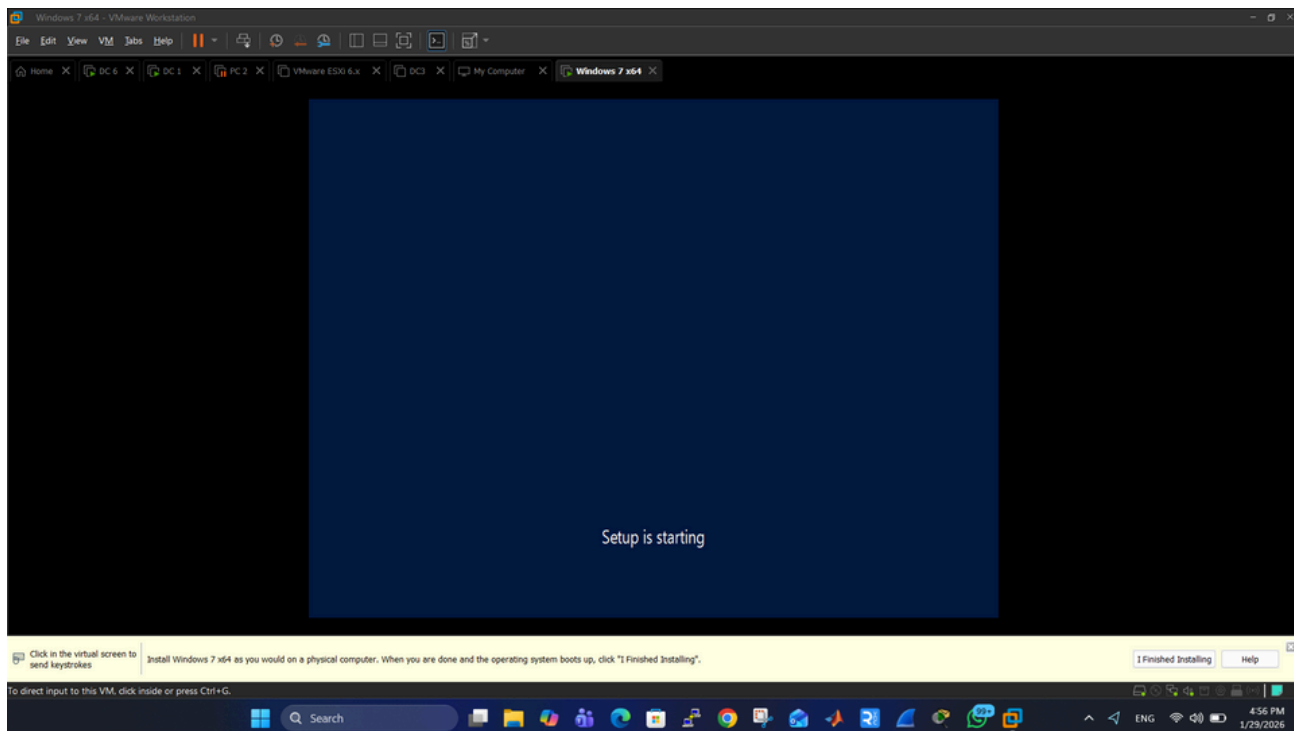
Goals:

- The WDS server is tasked with deploying Windows to 50 new computers that will be added to the ITI.Local domain.
- Once deployed, these computers will automatically join the domain, ensuring centralized management and policy enforcement.
- Alongside the hardware deployment, 50 new user accounts will be created in Active Directory. These users will be able to log in to the newly provisioned machines without additional configuration.

Benefits:

- **Efficiency:** Mass deployment reduces installation time compared to manual setup.
- **Scalability:** The WDS setup can be reused for future expansions, supporting additional computers and users.
- **Integration:** By joining the domain immediately after installation, the new computers inherit Group Policies, security settings, and organizational structures.

A PC boots directly from the network using PXE:



50 users created in primary domain controller:

